

Generative AI and Labor Market Outcomes: Evidence from the United Kingdom

Bouke Klein Teeselink^{*1,2,3}

¹*Department of Political Economy, King's College London*

²*AI Objectives Institute*

³*King's Institute for Artificial Intelligence*

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Abstract

This paper examines the effects of LLMs on UK labor market outcomes. We use a difference-in-differences design that compares outcomes across firms and occupations from 2021 to 2025 based on their differential exposure to LLM capabilities. Our results show that highly exposed firms reduce employment, with effects concentrated in junior positions. These firms sharply curtail new hiring, with technical and creative roles experiencing the steepest declines. At the occupation level, roles more exposed to LLMs also show substantial reductions in job listings. The observed effects concentrate almost entirely in high-wage segments. Hence, our findings suggest that LLMs may compress wage inequality.

Keywords: Generative AI; Large Language Models; Automation; Labor demand; United Kingdom.

JEL: O33; J23; J31

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sample construction by varying minimum firm size thresholds. We consider alternative task-level weights to construct our exposure measures. We estimate weighted regressions that give greater weight to larger firms and occupations to ensure our results reflect typical worker experiences. Finally, we relax the assumption of linear treatment effects by categorizing firms and occupations into exposure quartiles. These analyses consistently support our main findings.

These robustness checks notwithstanding, we acknowledge the possibility that other factors differentially affected exposed and less exposed firms or occupations around the time of ChatGPT’s introduction. For instance, if firms anticipating AI disruption had already begun adjusting their workforce composition before November 2022, or if other technological or economic shocks coincidentally affected high-exposure entities differently, our estimates could be biased. Additionally, our exposure measures capture potential susceptibility to LLM disruption but may not perfectly predict actual adoption patterns, which likely vary with firm-specific factors such as technological capabilities, management quality, and financial resources.

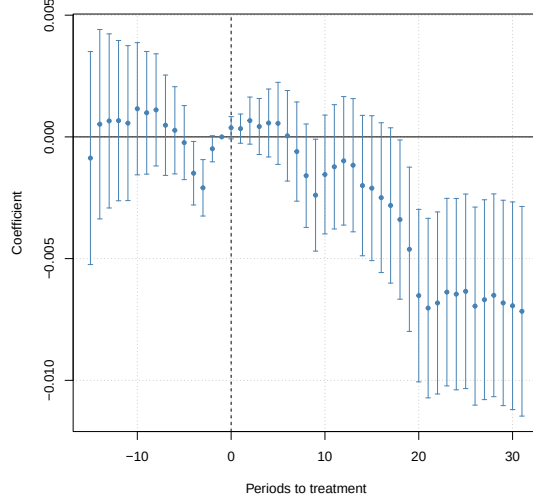
4 Results

This section presents our main empirical findings on the effects of LLM exposure on UK labor market outcomes. We organize the analysis in three parts. First, we examine how employment stocks, workforce composition, and compensation evolve within firms with different levels of LLM exposure. Second, we investigate occupation-level hiring patterns through job posting data. Third, we explore within-firm recruitment strategies by combining firm-level exposure measures with their occupation-specific hiring patterns. Throughout, we present both event study specifications and static difference-in-differences estimates.

4.1 Firm-Level Employment Effects

Figure 1 presents the event study plots for the effect of LLM exposure on the total number of employees in UK firms. We find that highly exposed firms start experiencing a statistically significant decline in total employment beginning approximately 18 months after the introduction of ChatGPT, suggesting that exposed firms reduce headcount in response to this new technology, but that it takes time to

Figure 1: Effect of LLM exposure on firm-level jobs



Notes: This figure plots event study estimates of the effect of LLM exposure on the total number of employees in UK firms (log). Coefficients are estimated from Equation (3). All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. Vertical line indicates November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

show up in aggregate employment statistics. High and low LLM exposure exhibit similar employment trajectories prior to November 2022, which provides support for the validity of the parallel trends assumption. When averaged over the entire post-treatment period (Table 3), a one standard deviation increase in LLM exposure is associated with a 0.3 percent reduction in total employment, although the aggregate effect is only marginally significant due to the delayed onset of impacts.

We next examine how these aggregate effects vary across different types of workers. Figure 2 decomposes the effects between high- and low-seniority employees. Panel A shows junior positions decline immediately and statistically significantly, while Panel B demonstrates that senior positions initially remain stable before declining gradually, with the decrease only becoming significant after 20 months. Panel C shows that the timing difference in the emergence of effects leads to a temporary shift toward senior employees, although the compositional change attenuates over time, with the seniority distribution gradually reverting toward pre-treatment levels by the end of our observation period. All three panels show parallel pre-trends.